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Team: Root Matrix

A02 Comparative Analysis Contribution

Individual comparison topics: TensorFlow vs. PyTorch; PyTorch vs. fastai

ITAI 2376 – Deep Learning in AI

Research and Analysis

I researched two assigned framework comparisons: TensorFlow vs. PyTorch and PyTorch vs. fastai. For each comparison, I examined the frameworks' origins and purposes, usability, performance and scalability, ecosystem and support, and real-world applications. I used official documentation, research papers, and framework case studies to verify technical claims and identify meaningful differences between the tools.

For TensorFlow vs. PyTorch, I analyzed how the frameworks developed from different priorities: TensorFlow's historical emphasis on scalable and heterogeneous deployment compared with PyTorch's flexible, Python-native development approach. I also examined how the two frameworks have converged over time as both have expanded their research, production, distributed-training, and accelerator capabilities. Real-world examples included TensorFlow applications and Amazon Ads' production use of PyTorch.

For PyTorch vs. fastai, I focused on their relationship rather than treating them as entirely separate competing frameworks. Because fastai is built on PyTorch, I analyzed the comparison primarily in terms of level of abstraction, developer productivity, and degree of control. I selected two medical-imaging examples to illustrate this distinction: fastai's chest X-ray/pneumothorax workflow and PathAI's custom PyTorch pathology system using multiple-instance learning. This allowed the comparison to demonstrate how fastai simplifies established workflows while direct PyTorch development provides greater flexibility for highly specialized applications.

Documents Produced

I wrote and formatted the following individual comparative-analysis reports:

- Comparative Analysis of TensorFlow vs. PyTorch
- Comparative Analysis of PyTorch vs. fastai

I edited the reports for clarity and conciseness, added supporting references, verified factual claims against source material, and included an AI-use disclosure.

Key Contribution to the Group Project

My primary contribution was the research, comparative analysis, source verification, and written documentation for these two framework pairs. I also developed a connecting theme between my two topics: the first comparison examines the choice between major deep-learning ecosystems, while the second examines the level of abstraction available within the PyTorch ecosystem.

Jupyter Notebook Demonstration

I created and tested a Jupyter notebook demonstrating the workflow differences among TensorFlow/Keras, PyTorch, and fastai using the MNIST dataset. I built comparable TensorFlow/Keras and PyTorch neural networks, trained and evaluated both models, and compared their accuracy, loss, and development workflows. I also created a fastai demonstration using a balanced MNIST subset to illustrate fastai's higher level of abstraction. I added interpretation notes and a final comparison table summarizing abstraction level, boilerplate, and developer control. During testing, I identified and corrected a class-imbalance issue in the initial fastai MNIST subset, then reran the demonstration with a balanced 5,000-image sample.